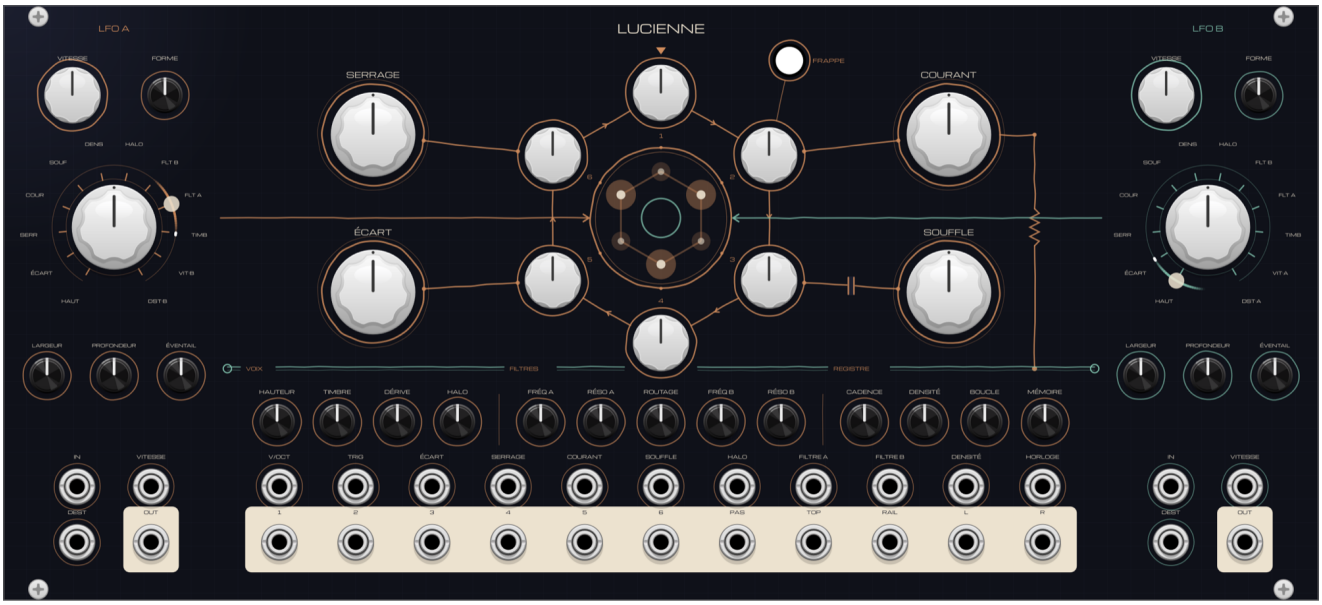
LUCIENNE

Six cores in a ring, tuned by ear, on a sagging power rail

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Lucienne is six analog-style oscillator cores tied in a ring. Each core is an integrator and a comparator, the circuit behind most triangle-and-square oscillators. You tune the six by ear, one knob each, above a common fundamental. SERRAGE then couples them: first they snap onto just intervals, then they modulate each other, then the ring turns chaotic.

COURANT starves their shared power supply: the pitch sags, the sound clips, voices cut out and come back. An internal random register opens and closes six low-pass gates; two resonant filters and a halo follow. Two slow, drifting LFOs move across a dial of twelve destinations.

The panel is in French. The tables below quote each label as printed, with its English sense the first time.

AT A GLANCE

- 6 triangle/square cores, each tuned from −12 to +36 semitones above a common fundamental
- SERRAGE: free, locked to just intervals, cross FM, chaos
- COURANT starves a shared rail: pitch sags, the sound clips, voices drop out
- Vactrol-style low-pass gates opened by an 8-step random register, or struck by TRIG
- Two resonant filters (24 dB ladder, 12 dB state-variable), series or parallel, then a halo
- Two drifting LFOs, each crossfaded across 12 destinations, including each other
- 56 HP

CONTROLS

Screen	The knot, in the middle of the six voices. Each voice is a dot that swells and brightens with its gate. The links of the ring thicken as SERRAGE tightens and tremble when the ring turns chaotic. The circle at the centre is the rail: it shrinks as it starves and flickers with the chaos.
1 to 6	Tuning of each voice, from −12 to +36 semitones above the fundamental set by HAUTEUR. The knobs are continuous: tune by ear. Voice 1, at the top, is the reference the others lock onto. At the start: 0, +7.02 (fifth), +12, +15.86, +19.02 and +26.04 semitones.
SERRAGE (grip)	How tightly the ring is coupled, 20 % at the start. At 0 the voices run free. Up to 50 % each voice is pulled more and more strongly onto the nearest just interval with voice 1, and its phase locks. From 35 % each voice is also frequency-modulated by the one before it in the ring, harder and harder, until the ring turns chaotic near 100 %.
ÉCART (stretch)	Scales the whole chord. 0: every voice on the fundamental (unison). 50 % (default): the chord as tuned. 100 %: every interval doubled. SERRAGE still pulls the result onto just intervals.
COURANT (current)	The voices' power supply, 100 % (clean) at the start. Turned down, the rail starts lower and sags more as the gates open: the pitch falls (about 3.5 semitones flat at 0 before any load, more under load), the voices clip harder, a 50 Hz hum and sudden drops creep in. It also drives the filters harder and damps their resonance. When the rail gets too low, voices cut out, voice 6 first and voice 1 last, and return when it recovers: the stack stutters.
SOUFFLE (breath)	Shape of the gates, from plucked to pad, 80 % at the start. Rise time from about 6 ms to 18 s (3.6 s at the start); fall from about 80 ms to 20 s, slowing as the gate closes, like a vactrol. Below 15 % the gates do not hold: each opening is a pluck. From 55 % they stay fully open as long as the register keeps them open.
FRAPPE (strike)	Strikes by hand: every voice the register keeps open restarts its attack. The light flashes. Same as a trigger at TRIG.
HAUTEUR (pitch)	The fundamental, from −1 to +3 octaves above C2: 32.7 Hz to 523 Hz. +0.5 (about 92 Hz) at the start. V/OCT adds to it; the filters follow it at half rate.
TIMBRE	Waveform of all six voices, 20 % at the start. 0: triangle. Around 45 %: square. Above: the triangle is folded, harder and harder up to 100 %. It also opens the gates' own low-pass a little.
DÉRIVE (drift)	Slow instability, 30 % at the start. Each voice wanders on its own, over a few seconds, typically a quarter of a semitone at 100 %; the whole module drifts together more slowly, about 12 cents at 100 %. 0: steady.
HALO	A diffuse, slowly modulated wash, like a dense reverb, 50 % at the start. 0: dry. Turning it up raises both its level and the length of its tail; at 100 % the dry sound is down to 55 %.
FRÉQ A	Cutoff of filter A, a saturating 24 dB/octave ladder low-pass, from 30 Hz to 16.5 kHz; about 4.7 kHz at the start. The value shown is exact with HAUTEUR at +0.5 and nothing in V/OCT. The right channel sits 3 % higher, for width.
RÉSO A	Resonance of filter A, 15 % at the start. Near 100 % it self-oscillates (less so on a starved rail).
ROUTAGE (routing)	How the filters are wired. 0 (default): series, A then B. 50 %: parallel, A and B mixed. 100 %: parallel, with B as a band-pass. It crossfades in between.
FRÉQ B	Cutoff of filter B, a 12 dB/octave state-variable filter, from 30 Hz to 16.5 kHz; about 8.8 kHz at the start. Same tracking as FRÉQ A.
RÉSO B	Resonance of filter B, 10 % at the start. It rings strongly at the top, kept round by a soft saturation.
CADENCE	Speed of the register, from 0.05 Hz (a step every 20 s) to 12.8 Hz; about 0.46 Hz at the start, a step every 2.2 s. A cable in HORLOGE replaces it.

DENSITÉ (density)	Share of the voices the register opens. 100 % (default): all six always open, a drone. 50 %: each voice open about half the steps. 0: all closed.
BOUCLE (loop)	Chance that the register takes a new random value on each step, 30 % at the start. 0: the same 8 values come round forever, an 8-step loop. 100 %: new values all the time.
MÉMOIRE (memory)	Lets the register move the pitches, 0 at the start. From about 25 % to 50 %, more and more voices jump an octave up or down on each step, so the chord stays just. Above 50 %, free offsets join in, up to ±3.5 semitones at 100 %. Each change glides in 6 ms.
VITESSE (rate)	LFO speed, from one cycle every 20 minutes to 30 Hz. At the start, A about 0.03 Hz (a 30 s cycle) and B about 0.09 Hz (11 s). Each LFO also wanders by a few percent on its own, so the two never stay in step.
FORME (shape)	LFO waveform, crossfading through sine (0), triangle (25 %), rising ramp (50 %), smoothed random (75 %) and stepped random (100 %).
Destination dial	The large knob ringed by the twelve marks (see the table). On a mark, the LFO goes to that destination; between two marks it is shared between both at constant power. At the start, A sits on FLT A and B on HAUT. The tooltip names the destination, or the pair. The arc shows where the LFO lands and how wide; the dot pulses with the LFO and dims as PROFONDEUR nears zero.
LARGEUR (width)	0 (default): the LFO reaches one mark, or the two either side of the pointer. Turned up, it spills onto more and more neighbours, weaker with distance; at 100 % it covers nearly the whole ring.
PROFONDEUR (depth)	Amount, from −100 % to +100 %; negative values invert. Gentle near the centre: 50 % gives 37.5 % of the full swing. At the start, A 25 % and B 12 %.
ÉVENTAIL (fan)	Phase offset from voice to voice on HAUT, ÉCART, DENS and TIMB: at 100 % the six voices are spread evenly over a whole cycle. On FLT A and FLT B it offsets the right channel from the left, up to half a cycle. Other destinations ignore it. At the start, A 30 % and B 60 %.

LFO DESTINATIONS

Mark	Destination	Full swing	ÉVENTAIL
HAUT	Pitch of each voice	±1.5 semitones	per voice
ÉCART	ÉCART	±20 %	per voice
SERR	SERRAGE	±35 %	—
COUR	COURANT	±45 %	—
SOUF	SOUFFLE	±40 %	—
DENS	DENSITÉ	±45 %	per voice
HALO	HALO	±35 %	—
FLT B	FRÉQ B	±30 % (about ±2.7 octaves)	left / right
FLT A	FRÉQ A	±30 % (about ±2.7 octaves)	left / right
TIMB	TIMBRE	±35 %	per voice
VIT·B · VIT·A	Speed of the other LFO	±3 octaves	—
DST·B · DST·A	Pointer of the other LFO	±3 marks	—

INPUTS

V/OCT	Added to HAUTEUR, 1 V per octave. 0 V leaves HAUTEUR where it is (it does not follow Rack's 0 V = C4). The filters follow at half rate. Lucienne is not a precision oscillator: DÉRIVE and the rail bend the pitch.
TRIG	While patched, the gates no longer hold: each rising edge (about 1 V) makes every voice the register keeps open rise and fall back, shaped by SOUFFLE. The register still picks which voices answer (DENSITÉ) and still drives MÉMOIRE.
ÉCART · SERRAGE · COURANT · SOUFFLE · HALO · DENSITÉ	Added to their knobs; 10 V covers the whole range.
FILTRE A · FILTRE B	Added to FRÉQ A and FRÉQ B; 10 V covers the whole range, about 1.1 V per octave.
HORLOGE (clock)	Replaces CADENCE: the register steps on each rising edge, above 1.5 V, ready again below 0.5 V.

OUTPUTS

1 to 6	Each voice on its own, after its gate and before the filters and the halo, ±5 V softly limited.
PAS (step)	The newest register value, 0 to 5 V, changing on each step: a stepped random voltage that loops when BOUCLE is low. It is the value voice 1 reads, so voice 1 is open while PAS is below DENSITÉ × 5 V.
TOP (tick)	A 10 V, 5 ms trigger on each register step, internal or from HORLOGE.
RAIL	The power rail, 0 to 10 V: 10 V when clean, lower as COURANT is turned down and as the gates open.
L · R	Stereo output, ±5 V. The voices are spread across the field, odd voices to the left, even voices to the right.

LFO JACKS

IN	Replaces the LFO with any signal: ±5 V for the full swing, up to ±7.5 V. VITESSE, FORME and ÉVENTAIL then have no effect; the dial, LARGEUR and PROFONDEUR still apply.
VITESSE	Added to the VITESSE knob, 1 V per octave.
DEST	Turns the dial: 10 V moves the pointer from the first mark to the last. Beyond, it wraps round.
OUT	The LFO itself, ±5 V, before PROFONDEUR and at voice 1's phase. A copy of IN when IN is patched.

HOW IT WORKS

Reading the destination table

Marks are listed clockwise, as on the dial. Full swing is the change with the LFO at its peak, PROFONDEUR at ±100 % and the pointer right on the mark; it adds to the knob and its CV input. On LFO A the last two marks read VIT·B and DST·B; on LFO B, VIT·A and DST·A.

Tuning by ear

Voice 1 is the reference. Every other voice measures its interval to voice 1 (after ÉCART, MÉMOIRE and DÉRIVE), folds it into one octave and looks for the nearest just interval: unison and octave, the fifth (3:2), the fourth (4:3), thirds and sixths (5:4, 6:5, 5:3, 8:5), then 7:4, 9:8, 15:8, 9:5, 7:5 and 16:15.

Simpler ratios catch from further away. With SERRAGE at 50 % or more, the octave reaches 0.8 semitone either side and 16:15 only 0.24. Inside half of that zone the voice snaps exactly; towards its edge it lets go gently. Outside, it stays where you tuned it, and you hear it beat against voice 1.

The voices also pull each other's phase: each gets a small kick whenever a neighbour flips, so a locked voice stays locked instead of sliding. Above 35 %, each voice's speed follows the triangle of the one before it (6 drives 1, 1 drives 2, and so on): cross FM, then chaos.

The power rail

The six voices share one supply. With COURANT at 100 % it sits at 10 V on the RAIL output and barely moves (at most about 25 cents of pitch with all six gates open). Turned down, it starts lower (5.5 V at 0) and sags much more as the gates open. Everything follows it: the pitch, the clipping inside each voice, the filters' drive and resonance.

Each voice has its own cut-out threshold, from about 4.5 V for voice 6 down to 3.3 V for voice 1. When a voice cuts out the load falls, the rail recovers and the voice returns: a starved patch stutters on its own. Some gain is given back as the rail drops, so the dying stack stays audible.

The register and the gates

The register holds 8 random values between 0 and 1. On each step they all move one place along; the oldest comes back as the newest unless BOUCLE replaces it. Voice N reads place N and is open while its value is below DENSITÉ. Since the values move along, a pattern travels round the voices, one voice per step.

On each step every open voice is struck again: low SOUFFLE plucks on every step, high SOUFFLE holds a drone. The gates are low-pass gates: closing, they get darker as well as quieter. Each Lucienne, and each Initialize, starts from its own random state, so two copies never play quite alike.

Two LFOs on one dial

Each LFO has a pointer on a ring of twelve marks, and each mark gets a share that falls with its distance from the pointer. Past the last mark the ring wraps round to the first (through the DEST input or the other LFO). The last two marks reach into the other LFO: A can speed up or slow down B, or swing B's pointer round the dial, and B can do the same to A. Two LFOs aimed at each other never settle.

PATCH IDEAS

1. Just drone

SERRAGE 12 %, SOUFFLE 90 %, HALO 60 %, DÉRIVE 35 %, TIMBRE 15 %, the rest at the start. Leave it running: the chord breathes with LFO A on filter A while LFO B makes the voices waver.

2. Tune a fifth by ear

SERRAGE at 0, voices 3 to 6 at 0. Turn voice 2 slowly around +7 semitones and listen to the beating. Raise SERRAGE to 40 %: as soon as it is close, it snaps onto the fifth and stays.

3. Brownout

SERRAGE 25 %, SOUFFLE 85 %, TIMBRE 35 %, HALO 40 %. Bring COURANT from 100 % down to 0 over a minute: the chord sags, hums, clips, then stutters as voices drop out. Watch RAIL on a scope.

4. Pinched knot

SOUFFLE 8 %, CADENCE 62 % (about 1.5 Hz), DENSITÉ 50 %, BOUCLE 15 %, MÉMOIRE 40 %, SERRAGE 60 %, TIMBRE 55 %, HAUTEUR +1.2, HALO 35 %: a plucked, looping figure that jumps octaves.

5. Vowel

ROUTAGE 50 %, FRÉQ A and FRÉQ B near the middle, RÉSO A 65 %, RÉSO B 70 %, TIMBRE 45 %. LFO A halfway between FLT B and FLT A, PROFONDEUR 80 %, ÉVENTAIL 50 %: the two peaks cross slowly, differently left and right.

6. Knotted LFOs

Both dials on VIT, LARGEUR 35 %, PROFONDEUR 70 %, VITESSE A 55 % and B 50 %. Each LFO speeds up and slows down the other, and spills onto TIMB and DST.

7. Struck

A clock every 1.5 s into TRIG, SERRAGE 35 %, RÉSO A 50 %. SOUFFLE 10 % for plucks, 50 % for swells that rise and fade.

8. A sequence on the side

PAS into a quantizer and on to another oscillator, TOP into the TRIG of an Ernest: the register plays them in time with the gates.

SPECIFICATIONS

Width	56 HP
Voices	6 integrator-comparator cores, 4× oversampled, each −12 to +36 semitones
Fundamental	32.7 Hz to 523 Hz (C1 to C5), plus V/OCT
Filters	A: 24 dB ladder low-pass. B: 12 dB state-variable, low-pass or band-pass. Both 30 Hz to 16.5 kHz.
Register	8 steps, 0.05 to 12.8 Hz or external clock
LFOs	2, one cycle every 20 minutes to 30 Hz (one an hour to 40 Hz with CV), 12 destinations
Levels	Audio and voice outputs ±5 V. PAS 0 to 5 V. TOP 10 V, 5 ms. RAIL 0 to 10 V. LFO OUT ±5 V. CV inputs: 10 V = full knob travel. TRIG about 1 V, HORLOGE 1.5 V.